

|            |                                                               |
|------------|---------------------------------------------------------------|
| Ex. Nos. 5 | Perform various configurations of Iptables Firewall in Linux. |
| Date :     |                                                               |

### ***Aim***

To perform and analyze various firewall configurations using IPTables in Kali Linux to control network traffic and enhance system security.

### ***Software Requirements***

- Host Operating System: Windows / Linux / macOS
- Virtualization Software: Oracle VirtualBox
- Guest Operating System: Kali Linux
- Security Tool: John the Ripper
- System Requirements:
  - o RAM: Minimum 4 GB (8 GB recommended)
  - o Disk Space: Minimum 30 GB
  - o Processor: Intel/AMD with Virtualization support

### ***Introduction***

IPTables is a powerful firewall utility in Linux used to configure rules for filtering incoming and outgoing network traffic. It operates by defining rules based on IP addresses, ports, and protocols. Configuring IPTables helps protect systems from unauthorized access, network attacks, and malicious traffic. Kali Linux provides built-in support for IPTables, making it suitable for learning firewall configurations in a controlled environment.

### ***Procedure***

#### ***Step 1: Start Kali Linux Virtual Machine***

1. Open Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

#### ***Step 2: Open Terminal and Check IPTables***

1. Open Terminal in Kali Linux.
2. Check IPTables version:
3. iptables --version

#### ***Step 3: View Existing Firewall Rules***

1. Display current IPTables rules:
2. sudo iptables -L

#### ***Step 4: Set Default Policies***

1. Set default policy to drop incoming traffic:
2. sudo iptables -P INPUT DROP
3. Allow outgoing traffic:
4. sudo iptables -P OUTPUT ACCEPT

#### ***Step 5: Allow Localhost Traffic***

1. Allow loopback interface traffic:
2. sudo iptables -A INPUT -i lo -j ACCEPT

### ***Step 6: Allow SSH Access***

1. Allow SSH connections on port 22:
2. `sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT`

### ***Step 7: Allow HTTP and HTTPS Traffic***

1. Allow HTTP traffic:
2. `sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT`
3. Allow HTTPS traffic:
4. `sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT`

### ***Step 8: Block a Specific IP Address***

1. Block traffic from a specific IP:
2. `sudo iptables -A INPUT -s 192.168.1.100 -j DROP`

### ***Step 9: Save IPTables Rules***

1. Save the configured rules:
2. `sudo iptables-save > firewall_rules.txt`

### ***Output***

Firewall rules are successfully applied.

Network traffic is filtered based on configured IPTables rules.

```
(student@kali)-[~]
$ iptables --version
iptables v1.8.4 (legacy)
(student@kali)-[~]
$ sudo iptables -L -n
Chain INPUT (policy ACCEPT)
target    prot opt source                destination
Chain FORWARD (policy ACCEPT)
target    prot opt source                destination
Chain OUTPUT (policy ACCEPT)
target    prot opt source                destination
(student@kali)-[~]
$ sudo iptables -F
$ sudo iptables -P INPUT DROP
$ sudo iptables -P OUTPUT ACCEPT
$ sudo iptables -A INPUT -i lo -j ACCEPT
$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
$ sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT
$ sudo iptables -A INPUT -s 192.168.1.100 -j DROP
$ sudo iptables-save > firewall_rules.txt
$ sudo iptables -L -n
Chain INPUT (policy DROP)
target    prot opt source                destination
ACCEPT   all  --  127.0.0.1              0.0.0.0/0
ACCEPT   tcp  --  0.0.0.0/0              0.0.0.0/0      tcp dpt:22
ACCEPT   tcp  --  0.0.0.0/0              0.0.0.0/0      tcp dpt:80
ACCEPT   tcp  --  0.0.0.0/0              0.0.0.0/0      tcp dpt:443
DROP     all  --  192.168.1.100          0.0.0.0/0
Chain FORWARD (policy ACCEPT)
target    prot opt source                destination
Chain OUTPUT (policy ACCEPT)
target    prot opt source                destination
(student@kali)-[~]
```

### ***Result***

Various firewall configurations were successfully implemented using IPTables in Kali Linux. The experiment demonstrates how firewall rules can be used to control network traffic and improve system security.